COMP2300

D1: Understanding the Problem Space

Carbon Footprint Awareness and Reduction Tool: India-Centric Challenge

23 October 2026 · Version 2.0

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AI-assisted consolidation and engineering work are disclosed in the accompanying genai.txt. Evidence limitations and work still requiring direct student confirmation are stated in the report; no unseen meetings, reviews or participant outcomes are asserted.

D1 — Literature review

From footprint estimates to feasible decisions

An India-focused tool must explain both the meaning of an estimate and the practical choices available to its user. Lee et al. [1] connect household consumption with supply-chain emissions and demonstrate substantial variation across Indian households and places. Their national analysis supports contextualisation but does not establish which interface helps an individual understand their footprint. A national average should therefore not be presented as a personalised baseline without explaining its scope.

Calculators can raise awareness without equipping users to act. Collins et al. [2] distinguish perceived learning and motivation from the information needed to change lifestyles. Their calculator-user evidence concerns reported perceptions, not a controlled demonstration of emissions reduction; people who choose a calculator may already be interested in sustainability. The design implication is to connect an estimate to a feasible decision, rather than assuming that displaying a total completes the job.

Hoffmann et al. [3] examine feedback across consumption domains with German university students. Their reductions concern emissions computed from planned consumption, not independently observed sustained behaviour. Effects differ by domain, so a promising overall result cannot justify a uniform intervention for mobility, food and household use. Neither the student sample nor its setting establishes transfer to Indian hostels, shift workers or multigenerational homes. A prototype can test comprehension and decision support first; sustained behaviour requires a different evaluation.

Feedback form, evidence and agency

Froehlich et al. [4] connect eco-feedback design with behavioural research and identify limited behavioural measurement in HCI work. An engaging interface is consequently not sufficient evidence of environmental benefit. For this project, evaluation should distinguish understanding, successful interaction, intended action and observed change. These outcomes require different measures and should not be collapsed into a single success claim.

Familiar equivalents also require testing. In hypothetical transport-choice studies, Mohanty et al. [5] found raw carbon dioxide by weight more effective than relatable equivalents for encouraging lower-carbon choices. This challenges the assumption that bus-trip or household comparisons automatically help. However, a hypothetical choice is not the same as accurate interpretation of a domestic estimate. The proposed comparison should retain the carbon unit and expose assumptions; a water bucket describes volume unless a defensible carbon conversion is supplied. Cost information similarly requires an explicit tariff and should not substitute for emissions.

Brynjarsdóttir et al. [6] challenge persuasive sustainability designs that reduce complex social circumstances to individual rational choices and simple metrics. This critique matters when transport safety, employment schedules and household infrastructure constrain action.

A technically correct estimate can still produce inappropriate advice. The design should preserve the estimated impact of a necessary activity while allowing users to reject unsuitable recommendations without explaining sensitive circumstances. Removing moral judgement must not mean silently deleting emissions.

Synthesis and research questions

The reviewed studies support contextualised estimates, transparent assumptions and actionable feedback, while leaving transfer and causal effects uncertain. This is a focused synthesis of six relevant works, not a systematic review or proof that no other research exists. The unresolved questions concern local interpretation, effort and shared responsibility. The accompanying coursework analysis examines these through S1–S3 and L1–L3; its illustrative interviews cannot establish population preferences.

RQ1: Do familiar comparisons and cost in INR improve understanding beyond carbon units alone? Compare interpretation accuracy, not preference alone. **RQ2:** Which presets reduce logging time/actions while preserving completeness and comprehension? Report the cost of setup and correction. **RQ3:** How can a shared-household view communicate communal totals without allocating them to individuals? Test understanding and attribution errors. Across all three, respect safety and schedule constraints. These questions turn literature gaps into assessable design decisions without promising carbon reduction.

References

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1. Scope, evidence and completed work

This report consolidates the existing literature review, problem-space analysis and prioritised epics for the confirmed India-centric carbon-footprint challenge. The literature review occupies the preceding two pages, including its references [1–6]. The rest of the report connects illustrative stakeholder evidence to six themes, six personas, four scenarios, one Job to Be Done and sixteen prioritised epics. The intended outcome is informed, feasible choice: no reduction in real emissions or household conflict has been measured.

The underlying CF1–CF10 interviews are described by the supplied analysis as illustrative coursework transcripts, rather than recorded field interviews. Original transcript files were not available for this consolidation. Coding summaries and quotations below are therefore reproduced or paraphrased from the supplied analysis [7], with its physical PDF pages identified. This preserves a usable evidence chain without pretending that the underlying transcripts have been rechecked. The existing JTBD and epic baseline is retained [8]; revised scope cautions are explicit. The official assessment requirements are taken from [9].

The present work consists of auditing and consolidating those documents, checking selected research sources, resolving ambiguous calculation labels, separating imagined scenario outcomes from observed results, and making traceability consistent. It does not establish new participant research or retrospective student meeting evidence.

2. Analysis method and traceability

The supplied analysis reports familiarisation, first-pass coding, clustering, checking candidate themes, defining patterns and writing the analysis [7, p1]. Hiranya has confirmed leading first-pass coding of CF5–CF7 and helping review the full set. Other detailed contribution assignments and historical meeting claims have not been independently confirmed in this revision. The table below preserves the existing coding evidence; it is not a claim of a new coding session.

A semantic theme describes what the accounts explicitly foreground. A latent theme interprets an underlying relationship, such as how billing arrangements distribute responsibility. Both must remain anchored to examples. Frequency is not used as a proxy for importance: a safety constraint can justify a design requirement even when it appears in few accounts. Conversely, a vivid quotation cannot establish prevalence or causality. Disagreement and alternative explanations are retained where they affect design.

First-pass code and theme matrix

CF1: Pune hostel student

Retained first-pass codes: Curfew cab; unfamiliar carbon units; exam delivery plastic; shared geyser; small data allowance; Hindi/Marathi

Pattern / theme links: Legibility and effort; communal loads; safety constraint → S1, S2, S3, L1, L3

CF2: Bengaluru hybrid engineer

Retained first-pass codes: WFH cooling; opaque bill units; city-matched comparison; packaging and fast fashion; dislikes lectures

Pattern / theme links: Contextual estimates; consumption beyond transport; feasible advice → S1, S2, S3, L3

CF3: Noida household manager

Retained first-pass codes: Repeated reminders; different family comfort needs; grocery spoilage; composting; tanker water; shared Hindi view

Pattern / theme links: Unequal coordination burden; visible everyday waste → S1, S2, S3, L2

CF4: Chennai postgraduate

Retained first-pass codes: Bicycle and repair; doubts foreign factors; village/city fuel differences; inspect assumptions; simple/detailed views

Pattern / theme links: Trust, recognition and inspectability → S1, S2, S3, L3

CF5: Thane commuter

Retained first-pass codes: Train with rain alternatives; packaging guilt; online van versus market uncertainty; couple disagreements; easiest swap

Pattern / theme links: Convenience trade-offs and negotiated action → S1, S2, L2, L3

CF6: Gurugram household

Retained first-pass codes: Long car commute; school run; family Metro use; AC; rupees and kilograms; child's questions

Pattern / theme links: Comparable options and shared explanation → S1, S2, L2

CF7: Kolkata shared-flat student

Retained first-pass codes: Landlord-paid electricity; food leftovers; morning bucket water; minimal typing; weekly picture; Bengali

Pattern / theme links: Cost/use separation and low-effort access → S2, S3, L1

CF8: Lucknow elder

Retained first-pass codes: Fan and bucket habits; repair/composting; plain Hindi; help from grandson; habits dismissed as old-fashioned

Pattern / theme links: Familiar explanations and intergenerational respect → S1, S3, L2

CF9: Hyderabad shift worker

Retained first-pass codes: Cab after buses stop; daytime sleep cooling; exhaustion; tiffin when feasible; rota and quiet notifications

Pattern / theme links: Structural constraints and attention protection → S2, S3, L3

CF10: Ahmedabad co-living resident

Retained first-pass codes: Equal bill split; common AC/geyser; phone and clothing replacement; one monthly challenge

Pattern / theme links: Diffused responsibility and overlooked consumption → S2, S3, L1

Source: supplied coding table [7, pp1–4]. These summaries are analytical anchors, not population statistics. CF5–CF7 are Hiranya's confirmed initial lead batch; full-set review prevents the batch boundary becoming the theme boundary.

3. Semantic themes

S1 — Carbon needs a meaningful, inspectable context

The pattern is a gap between the quantity shown and the meaning a person can use. CF1 cannot picture carbon units; CF6 asks for financial and carbon information together; CF4 wants the assumptions exposed; CF2 wants a relevant local comparison. The issue is therefore more than translating terminology: trust depends on units, geography, scope and the relationship to a decision.

Illustrative quotation: “Show me in rupees and kilos, both.” (CF6, reproduced in [7, p4]). A second example is “A unit of electricity here is much dirtier. We’re still heavily on coal... I’d want to see the assumptions.” (CF4, [7, p4]). These are stakeholder statements, not verified emission-factor measurements.

Lee et al. [1] support contextual variation; Mohanty et al. [5] complicate the proposed response because raw CO2 outperformed equivalents in their hypothetical transport-choice setting. A request for familiar units is evidence of preference, not proof of comprehension. The design implication is to retain the carbon metric, show assumptions and test comparison formats. Water buckets measure volume; INR measures cost; neither is a carbon equivalent without a valid model. RQ1 tests this distinction. E1 and E2 address legibility and credibility; E13 remains conditional on suitable benchmark data.

S2 — Convenience and replacement are visible; existing care is easily overlooked

Delivery packaging, spoilt groceries, clothing replacement and phone upgrades recur across different contexts. Repair, reuse and composting provide a counter-pattern: a tool focused only on purchases could ignore practices users already value. CF5's question about the delivery van also prevents the simplistic conclusion that shopping locally must always be better.

Illustrative quotation: “I genuinely want to know if the market is better than ordering online, because the van is a factor too.” (CF5, [7, p5]). CF4 adds: “If the tool only counts what you buy, it misses all that.” ([7, p5], referring to repair and passing items on).

The implication is to expose category coverage and acknowledge existing habits. It is not to invent avoided-emission credits. A repair record can be meaningful without a numerical offset; transport, product lifetime and disposal assumptions would be needed to quantify one. E4 preserves category visibility; E8 recognises practices; E16 excludes unsourced credits. The tension between recognition and defensible calculation is a deliberate design boundary.

S3 — Attention, language and effort govern access

Several accounts reject long forms and repeated typing while asking for short summaries, familiar language or optional depth. CF4 wants inspection and export; CF7 prefers a weekly picture; CF8 needs plain Hindi and assistance. These needs challenge a single interface that assumes every user has the same time, confidence or vocabulary.

Illustrative quotation: “A simple weekly picture would be better than a long report. Oh, and it should be in Bengali too, my roommate doesn’t read English well.” (CF7, [7, p5]). The account expresses an access need, not a demonstrated usability result.

Presets and progressive disclosure are plausible responses, but fewer taps can conceal defaults and errors. RQ2 must therefore measure completeness and comprehension as well as time. The initial language set requires an explicit decision and actual copy review; a language toggle alone is insufficient. E5 prioritises offline, low-effort entry. E12 separates optional depth from the basic journey. Notifications remain optional because attention and rest are costs, not free resources.

4. Latent themes

L1 — Separating payment from use can diffuse responsibility

CF1's shared geyser, CF7's landlord-paid electricity and CF10's equal bill split connect individual action weakly to personal payment. The latent interpretation is that feedback must help a group understand a shared resource without assigning a culprit. It would be too strong to conclude that people do not care: unclear responsibility and lack of a trusted collective view are alternative explanations.

Illustrative quotation: “The landlord pays the electricity, so none of us really think about it.” (CF7, [7, p6]). The account is evidence for this interpretation within the supplied material, not proof that all tenants behave similarly.

E6 therefore defaults to communal totals, while E7 supports optional manual sharing. A shared bill cannot reveal individual consumption. Even consent does not create measurement: any optional individual view needs both valid attribution and that person's own consent. RQ3 tests whether the interface communicates the boundary. This is distinct from S3's interaction effort and L2's unequal family coordination.

L2 — Conservation work is also coordination work

CF3 describes repeatedly reminding family members; CF5 describes disagreement within a couple; CF8's low-waste practices are dismissed as old-fashioned, while younger relatives can help him use technology. The latent pattern is unequal responsibility for noticing, explaining and coordinating consumption. It is not simply a stereotype of older versus younger people: children also raise questions, and assistance can be reciprocal.

Illustrative quotation: “I do. Nobody else. Lights off when you leave a room, I’m always saying it...” (CF3, [7, p6]). CF5's “We’re on the same team, but we don’t always agree on what matters.” ([7, p6]) makes disagreement visible without casting a person as the problem.

A shared explanatory view could support discussion, but could also amplify surveillance or one person's authority. The proposed response is household totals, accessible wording and voluntary participation; no public rankings or pressure after declining attribution. P5 and P6 embody distinct needs within the same design scenario. E6, E7 and E8 respond to coordination and recognition. Reduced conflict remains a hypothesis to evaluate, not a promised outcome.

L3 — Some high-impact choices are constrained rather than discretionary

Curfew travel, unavailable late-night buses, shift work and necessary cooling connect behaviour to conditions a user cannot readily change. The interpretation is that recommendations should consider feasibility before ranking environmental benefit. CF5's fatigue and CF2's rejection of lectures broaden the issue beyond a single occupation.

Illustrative quotation: “Night shift I take a cab, because the buses have stopped... Work around my rota... please, no notifications at night. I’m trying to sleep.” (CF9, [7, p6]). The ellipses are retained from the supplied analysis; they are not presented as an unabridged transcript excerpt.

This theme accords with the critique of narrowly individual persuasion [6] and with the gap between feedback and action [2–4]. E3 lets a user mark a necessary activity without providing a sensitive explanation. Its impact remains visible, but alternatives requiring that activity to change are excluded. E11 couples any later notification with opt-in and quiet hours. The principle is feasible action without blame, not erasing inconvenient emissions.

5. Synthesis, tensions and design hypotheses

Together, the themes suggest three connected obstacles: interpreting a limited estimate, finding a feasible response, and negotiating a shared context. A calculator that only totals entries addresses none of these reliably. The working design hypotheses are that transparent assumptions, low-effort entry and a constraint-aware action path can improve understanding and decision support. Evidence does not yet establish retention, real-world savings or representative Indian demand.

Familiar versus accurate

Evidence and interpretation: S1 requests equivalents; [5] favours raw CO2 in a different task

Design consequence and unresolved question: Keep units explicit and test comprehension; do not assume the preferred representation wins

Recognition versus rigour

Evidence and interpretation: S2 values repair; no approved avoided-emissions method exists

Design consequence and unresolved question: Acknowledge descriptively; numerical credits remain excluded

Shared totals versus “own part”

Evidence and interpretation: L1/L2 support communal visibility; CF3 wants members to see their part

Design consequence and unresolved question: Default communal view; optional attribution requires measurement and enforceable consent

Fast entry versus trustworthy estimates

Evidence and interpretation: S3 rejects forms; missing details change calculations

Design consequence and unresolved question: Presets remain visible/editable; unknown remains unknown; test correction costs

Action versus constraint

Evidence and interpretation: L3 limits apparent alternatives

Design consequence and unresolved question: Retain necessary activities, filter unsuitable advice, allow neutral no-option state

Rich scope versus delivery capacity

Evidence and interpretation: Broad everyday categories and language needs

Design consequence and unresolved question: Choose an explicit quantitative subset later; disclose descriptive/omitted categories

The research questions remain those of the preceding review: RQ1 interpretation of quantities and cost; RQ2 low-effort entry without loss of meaning; RQ3 communal understanding without unsupported individual attribution. Safety and schedule constraints are a cross-cutting evaluation condition. D2 must resolve consent design, category coverage, language scope and the precise entry-effort targets before those can become implementation acceptance criteria.

6. Personas

Personas are synthetic design constructs, not additional participants. Details below preserve the supplied baseline [7, pp9–12]. Composite relocations and relationships are explicitly identified; they must not be mistaken for transcript facts. Three stakeholder groups are represented: students/shared housing (P1–P2), time-poor professionals (P3–P4), and household managers/assisted elders (P5–P6).

P1 — Ananya Deshmukh: hostel student

Basis: CF1, with CF7 informing shared-housing and language access; S1–S3, L1, L3. A 21-year-old engineering student in Pune, Ananya usually walks or uses public transport but needs a cab for late journeys before hostel curfew. Exam-time delivery packaging and a shared geyser are salient concerns. A small data allowance and limited patience for forms make offline entry important.

Goals and motivation: picture relevant activities, record a manageable amount, and discuss shared use without an argument. **Frustrations:** unfamiliar carbon units, hidden shared costs and recommendations that ignore safety. **Design relevance:** a short optional setup, editable presets, explicit category coverage and a necessary-trip flag. The goal is intelligible choice, not a score for compliance. Primary scenario SC2; E1–E7 and optional E10.

P2 — Karthik Sundaram: postgraduate and repair advocate

Basis: CF4; S1–S3 and L3. A 24-year-old environmental-science postgraduate in Chennai with rural Tamil Nadu experience, Karthik cycles, uses alternatives in rain and repairs possessions. He speaks Tamil, Hindi and English and uses a laptop for research.

Goals and motivation: inspect geographic and calculation assumptions, recognise repair practices, and access detail without imposing it on novice users. **Frustrations:** unexplained foreign defaults and a purchase-only account of sustainability. **Design relevance:** inspectable factors, an optional audit/export view, and descriptive recognition when credits

cannot be justified. His scepticism does not validate a particular factor by itself. Primary scenario SC4; E2, E4, E6, E8 and E12.

P3 — Rohan Kulkarni: hybrid worker in co-living

Basis: CF2 and CF10; S1–S3, L1 and L3. The baseline combines Bengaluru work/cooling habits with Ahmedabad co-living and replacement habits and relocates this composite to Chennai to share SC4 with P2. This relocation is a narrative device. Rohan is 29, works partly at home, runs cooling and computer equipment, shares bills and dislikes repeated logging.

Goals and motivation: understand one honest period estimate and consider a manageable alternative or monthly challenge. **Frustrations:** opaque bills, packaging and equal bill splits that hide differences. **Design relevance:** transparent what-if comparisons and optional depth; a reference-home benchmark only if a matching source exists. No inference of room consumption from a shared bill. Primary scenario SC4; E1–E6, E9–E10 and conditional E13.

P4 — Priya Reddy: rotating-shift professional

Basis: CF9 with CF5 informing feasible-action framing; S2, S3 and L3. Priya is a 31-year-old Hyderabad nurse whose travel and daytime sleep cooling depend on shift patterns. She sometimes brings a steel tiffin but cannot assume time or energy to cook after every shift.

Goals and motivation: useful advice around her actual constraints and short, optional feedback. **Frustrations:** unsafe transport suggestions, moralising scores and notifications during rest. **Design relevance:** a generic necessary-use flag, dismissal of unsuitable suggestions and no request for health information. Quiet hours are inseparable from any future notification. Primary scenario SC3; E3–E5 and optional E11.

P5 — Sunita Sharma: household coordinator

Basis: CF3, with CF5–CF6 as secondary context; S1–S3 and L2. The baseline places Sunita, 46, in a Noida multigenerational household. She manages resource use, notices grocery spoilage and handles composting and recyclables. Her family relationship with P6 is constructed for SC1.

Goals and motivation: make communal use understandable without carrying every reminder herself; explain water and electricity in familiar terms. **Frustrations:** conflicting comfort needs, ignored reminders and waste that does not appear on a bill. **Design relevance:** shared totals, clear volume/cost labels, explicit exclusions and optional household cards. A request to see individual shares is constrained by consent and measurement, not automatically satisfied. Primary scenario SC1; E1, E4–E9 where appropriate.

P6 — Rajeshwar Prasad Verma: assisted elder

Basis: CF8, with CF3 as family context; S1, S3 and L2. The baseline relocates CF8's Lucknow context to Noida and makes Rajeshwar, 58, Sunita's father-in-law. This relationship is synthetic. He values fan use, bucket watering and composting, prefers plain Hindi and may ask grandchildren for help.

Goals and motivation: understand familiar quantities and have existing habits respected. **Frustrations:** complicated menus, jargon and dismissive treatment of frugal practices. **Design relevance:** readable labels, a simple default, assisted use without surrendering control and no invented savings credit. Primary scenario SC1; E1, E2, E5, E6 and E8. Age alone is not used to infer inability.

7. Goal-driven scenarios

These scenarios illustrate proposed use, not observed outcomes. Their success conditions state what later prototypes should enable. Numerical examples are synthetic arithmetic fixtures. They do not establish local tariffs, appliance consumption or an official emissions factor. Unsupported bus-trip carbon comparisons from the earlier draft are omitted because the denominator and service boundary have not been justified.

SC1 — Shared water and electricity in a Noida household

Sunita wants to discuss a teenager's long shower while Rajeshwar wants a plain explanation of water use. They open a communal view and compare quantities without assigning blame. In an illustrative example, 20 minutes at an assumed 7.5 litres per minute equals 150 litres, or ten assumed 15-litre buckets. The interface labels this as volume, not carbon. A fan costing illustration uses 75 W for one hour and an assumed INR 8/kWh: 0.075 kWh and INR 0.60, excluding fixed charges and taxes.

The user can inspect or change assumptions. Shared totals remain communal; no person's share appears merely because Sunita requests it. Existing bucket or composting habits receive acknowledgement without an offset. **Success condition:** both users can explain the comparison and its limitations. Shorter showers or reduced conflict are potential outcomes, not established results. P5/P6 and indirect family readers; S1–S3/L2; E1, E2, E4–E6, with E7/E8 optional.

SC2 — Exam-time activity and a shared geyser in Pune

Ananya has little time and mobile data during exams. She records packaging with an editable offline preset and marks her late cab as necessary. The journey remains visible, while advice does not require unsafe travel. She examines a shared geyser: an assumed constant 2 kW for two hours gives 4 kWh. Actual thermostatic cycling is unknown, so this is a model, not measured waste.

A carbon figure is unavailable until a compatible factor is supplied; no invented bus-trip equivalent is shown. A communal summary could later be shared manually in reviewed languages. **Success condition:** Ananya can record the activity, distinguish missing from

zero data, and see one feasible option without safety blame. Two-tap or five-second performance is a target to test, not a result. P1 and floor-mates; S1–S3/L1/L3; E1–E6, E7 optional.

SC3 — Rest and necessary travel in Hyderabad

Priya records a necessary late-night cab and necessary daytime cooling without entering a diagnosis or explanation. The tool excludes suggestions to change those records and offers a different category only if feasible. She may reject a tiffin suggestion when preparation is impractical; a neutral no-option state is preferable to repeated pressure.

If a later notification feature is enabled, it must be opt-in, respect editable quiet hours and send no more than the agreed frequency. The initial journey works without notifications. **Success condition:** Priya can understand the remaining estimate and dismiss unsuitable advice without penalty or interrupted rest. No reduction in packaging is claimed. P4; S2/S3/L3; E3–E5, E11 optional.

SC4 — Inspecting a co-living estimate in Chennai

Rohan and Karthik inspect a shared electricity record. A meter total and an appliance breakdown must not be added together when the latter is already included. Karthik inspects the factor and its evidence status; Rohan previews changing eligible cooling from eight to seven hours per day across fourteen days. With a synthetic constant power of 1.5 kW, the difference is 21 kWh. At the illustrative factor 0.716 kg CO2/kWh and INR 8/kWh, that gives 15.036 kg CO2 and INR 168 of estimated variable cost.

The factor is explicitly illustrative and CO2 is not relabelled CO2e. Choosing the action records an intention, not a saving. Phone repair is acknowledged without a numerical credit. Export, benchmarks and monthly challenges remain optional extensions. Any smart-plug timer is configured outside the proposed tool, not represented as implemented integration. **Success condition:** both users can explain inputs, overlap and the conditional comparison. P2/P3; S1–S3/L1/L3; E1, E2, E4–E6, with E8–E10/E12 optional and E13 conditional.

8. Job to Be Done

When I want to understand how my everyday life affects the climate, I want to see which activities matter most and which realistic alternatives are open to me, so that I can make changes that fit my budget, safety, schedule and household without being misled or blamed for constraints I cannot control.

This 52-word statement preserves the supplied baseline [8, p1]. It expresses an outcome rather than specifying an interface. Understanding responds to S1/S2, feasible alternatives to L3, household fit to L1/L2, and avoiding misleading feedback to S1/S3. It stops at informed choice because the evidence does not establish sustained behaviour change. Each scenario tests a different obstacle to that shared job rather than inventing four unrelated products.

9. Epics and MoSCoW prioritisation

A Must supports the core journey in at least two scenarios, or its omission would make the tool misleading, exclusionary or harmful. A Should adds depth, communication or repeated-use value while the basic journey remains possible. A Could depends on unsettled data or additional capacity. A Won't is excluded from this version with a stated revisit condition. These are value/risk judgements from illustrative evidence, not measured demand or a numerical priority score.

Must have

E1 — Understand a footprint in familiar terms. As a newcomer, I want a manageable selection of everyday activities explained in relevant units so I can understand what matters. P1/P3/P5/P6; S1, RQ1; SC1/SC2/SC4. Show quantities and, where justified, INR and a compatible carbon unit. No mandatory unknown answer becomes zero. Volume and cost remain distinct from carbon. Without this, the core job of understanding fails.

E2 — Inspect the working. As a sceptical user, I want to inspect and correct assumptions so I can judge credibility. P2/P3/P6; S1/S3; SC1/SC2/SC4. Expose value, unit, source, year, geography and evidence status; disclose user overrides and preserve the version behind an estimate. Unknown factors produce descriptive results. This is Must because an unexplained number can mislead even when arithmetic is correct.

E3 — Find feasible actions. As a constrained user, I want advice that respects safety, health, time and housing so I can choose something achievable. P1/P4; L3/S3; SC2/SC3. Necessary activities retain their impact but are not treated as discretionary. Allow dismissal and a neutral no-option state. Omission could produce harmful or alienating advice.

E4 — See categories and gaps. As a user, I want relevant activities visible and missing coverage explicit so I do not mistake a subtotal for a whole footprint. P1–P5; S2; SC1–SC4. Keep food, packaging, clothing, devices, water and waste visible where appropriate, but quantify only where methods are defensible. A descriptive record is preferable to an invented estimate.

E5 — Record briefly and offline. As a busy or language-constrained user, I want a short, understandable entry path so I can participate. P1/P4/P5/P6; S3/RQ2; SC1–SC3. Proposed targets are a short setup and editable presets. Exact actions, time and language scope require D2 decisions and tests. English/Plain Hindi is an initial proposal, not proof of support for every language requested.

E6 — See shared use without blame. As a household member, I want communal totals by default and control over any optional own-share view. P1/P2/P3/P5/P6; L1/L2/RQ3; SC1/SC2/SC4. No member can opt in for another; decline produces no pressure. Optional shares require valid measurement as well as consent. This Must is explicitly unresolved in part: a communal-only implementation does not fulfil the entire optional-share requirement. D2 must record a defensible scope decision.

Should have

E7 — Share a household card manually. P1/P5 and indirect readers; L1/L2; SC1/SC2. The user previews communal coverage and controls export; no automatic transmission or unconsented individual detail. Sharing strengthens discussion but is not required to understand an estimate.

E8 — Recognise repair and low-waste habits. P2/P5/P6; S2/L2; SC1/SC4. Record and acknowledge practices; only a justified method can create a numeric avoided-emissions figure. This adds recognition without making unsupported credits essential.

E9 — Compare options and what-ifs. P3/P5; S1/L3; SC4. Compare equal periods, scope and assumptions, distinguishing cost, impact and feasibility. General comparison depth is Should; D2 may refine a narrow action preview as necessary for E3. Commute and delivery comparisons need additional scenarios and methods before implementation.

E10 — Try one challenge and look back. P1/P3; S3; SC4. One voluntary challenge at a time; compare compatible observations and separate intent from achievement. No streak or retention claim is justified by the current evidence.

E11 — Receive quiet, optional feedback. P4; L3/S3; SC3. If notifications are introduced, include opt-in, editable quiet hours and limited frequency together. Rota entry effort must be tested; no unverified two-tap promise is retained. The core journey works without alerts.

E12 — Inspect deeply and export own records. P2; S3; SC4. Keep a simple default while exposing factor provenance and calculation detail. CSV export is a proposed second-increment feature and must avoid leaking another person's data. Export supports auditing but is not needed for the first explanation.

Could have

E13 — Compare with a relevant reference home. P3; S1; SC4. Use a sourced table matching region, period and household definition; explain boundaries and keep it opt-in. Hide the feature without appropriate reference data. A national mean is not automatically an appropriate personal target.

Won't have in this version

E14 — Automatic device/account data collection. S3/P3. Revisit when a real data source, consent and privacy handling are established. The official topic lists smart-device/app integration without explicit priority; this exclusion needs supervisor clarification rather than being treated as a waived official requirement.

E15 — Public rankings or attribution without consent. L1/L2/L3. Unequal constraints undermine rankings, and a shared bill cannot reveal individual use. Revisit individual attribution only with measurement and the relevant person's consent; neither is inferred from household membership.

E16 — Unsourced avoided-emission credits. S2/P2/P5/P6. Repair, recycling and composting remain descriptively recognisable under E8. Numerical credits require a defensible method, system boundary and transparent assumptions before reconsideration.

10. Traceability and downstream decisions

SC1: household water/fan

Core epics: E1, E2, E4, E5, E6

Supporting scope and unresolved issues: E7/E8 optional; no E16 credit; reviewed language and own-share consent needed

SC2: hostel and protected travel

Core epics: E1–E6

Supporting scope and unresolved issues: E7 optional; no assumed bus factor or achieved five-second logging

SC3: shift constraints

Core epics: E3, E4, E5

Supporting scope and unresolved issues: E11 optional; no notifications until quiet-hour protection exists

SC4: audit and eligible comparison

Core epics: E1, E2, E4, E5, E6

Supporting scope and unresolved issues: E8–E10/E12 optional; E13 data-dependent; no automatic integration claim

Every persona has a core route: P1 through SC2; P2/P3 through SC4; P4 through SC3; P5/P6 through SC1. Indirect stakeholders include floor-mates, family members and assisted readers, whose privacy and comprehension matter even if they do not enter data. The table is an evidence-to-design map, not test completion evidence.

D2 should compare distinct concepts before committing to an interface; split epics into small stories with observable acceptance criteria; resolve the quantitative category subset, consent mechanism and language scope; and test the difference between preferred and understood representations. The Java/JavaFX requirement is an implementation constraint, not a reason to silently promise a phone app. No speculative machine learning is needed to address the present job.

11. Project process, tools and critical reflection

Work summary and task report. The supplied artefacts contain the research synthesis, a CF1–CF10 coding matrix, six themes, six personas, four scenarios and an epic baseline. This revision consolidates them into a single report, checks six selected research sources, standardises traceability and corrects unsupported numerical or outcome framing. Hiranya's confirmed contribution is first-pass coding of CF5–CF7 and review of the full set. The role preferences supplied for Arnav and Keshav describe tendencies, not independently verified task completion.

Management and tools. The present revision used a source inventory, requirement audit, risk ordering and bounded work allocation between AI drafting/review tasks. Markdown provides editable text; Python and ReportLab provide reproducible document generation; PDF text extraction and rendering support checking. AI assistance is disclosed separately. Earlier claims of twice-weekly meetings, universal student peer review and completed Scrum ceremonies are not repeated without actual records. The official weekly minutes and PRISM requirements still apply; they cannot be satisfied by this generated report.

Product reflection. The strongest aspect of the baseline is its chain from everyday constraints to a non-moralising JTBD. Its main weaknesses were inconsistent evidence labels, unverified factor placeholders, overly certain scenario outcomes and the tension between communal totals and optional individual shares. The revision replaces placeholder certainty with explicit limitations, retains source identifiers and states the consent conflict. This makes the project more defensible, although original transcripts are still needed to check the quotations and coding directly.

Process reflection. Separate documents allowed citation numbering, locations and scope statements to drift. The concrete improvement is a shared traceability register linking source, code, theme, persona, scenario and epic, with one identifier set and explicit proposed/verified states. The next review should challenge negative cases and rejected alternatives, not just formatting. A polished PDF is only the presentation layer of an evidence chain; it cannot compensate for undocumented contribution or unavailable source verification.

12. Limitations and readiness

The supplied interviews are illustrative and the original transcripts were unavailable for this consolidation. Personas and scenarios are constructed; no population estimate, usability result or behavioural effect follows from them. The literature review is focused rather than systematic. Numerical examples are model fixtures, not approved defaults. Own-share consent, languages, benchmarks and integration scope remain decisions for subsequent work.

The revised content addresses every D1 content area, but a full submission claim still requires the students to verify quotations against the original transcripts, confirm the team/process record, review the final PDF and supply the required disclosure. Meeting notes

and individual reflections must remain truthful and separately submitted where required. These limitations are material, not hidden in a claim of guaranteed distinction.

References for the integrated report

References [1–6] are repeated here from the self-contained two-page literature review so that the integrated report ends with a complete reference list.

[1] J. Lee, O. Taherzadeh, and K. Kanemoto. 2021. The scale and drivers of carbon footprints in households, cities and regions across India. *Global Environmental Change* 66, 102205. https://doi.org/10.1016/j.gloenvcha.2020.102205

[2] A. Collins, A. Galli, T. Hipwood, and A. Murthy. 2020. Living within a One Planet reality: The contribution of personal Footprint calculators. *Environmental Research Letters* 15, 025008. https://doi.org/10.1088/1748-9326/ab5f96

[3] S. Hoffmann, W. Lasarov, H. Reimers, and M. Trabandt. 2024. Carbon footprint tracking apps. Does feedback help reduce carbon emissions? *Journal of Cleaner Production* 434, 139981. https://doi.org/10.1016/j.jclepro.2023.139981

[4] J. Froehlich, L. Findlater, and J. Landay. 2010. The design of eco-feedback technology. In *CHI '10*, 1999–2008. ACM. https://doi.org/10.1145/1753326.1753629

[5] V. Mohanty, A. Filipowicz, N. Bravo, S. Carter, and D. A. Shamma. 2023. Save a tree or 6 kg of CO_2? Understanding effective carbon footprint interventions for eco-friendly vehicular choices. In *CHI '23*. ACM. https://doi.org/10.1145/3544548.3580675

[6] H. Brynjarsdóttir, M. Håkansson, J. Pierce, E. Baumer, C. DiSalvo, and P. Sengers. 2012. Sustainably unpersuaded: How persuasion narrows our vision of sustainability. In *CHI '12*, 947–956. ACM. https://doi.org/10.1145/2207676.2208539

[7] Arnav Mishra, Hiranya Makkar, and Keshav Singhal. 2026. *Deliverable 1: Interview Analysis, Personas and Scenarios*. Version 1.7, October 2026. COMP2300 working document, University of Southampton Delhi. Supplied file: ThemAnalysis.pdf. CF1–CF10 quotations and coding are cited indirectly through this document because the original transcripts were unavailable.

[8] Arnav Mishra, Hiranya Makkar, and Keshav Singhal. 2026. *Deliverable 1: Job to Be Done, Epics and MoSCoW Prioritisation*. Version 1.1, October 2026. COMP2300 working document, University of Southampton Delhi. Supplied file: JTBD Epics MoSCoW.pdf.

[9] COMP2300 Module Team. 2026. *Software Design and Development Project: Coursework Specification 26/27*. Draft version 1, 23 September 2026. University of Southampton. Supplied file: Coursework Specificartions.pdf, pp9–12, 30–40.